

PrimeSim™ Reliability Analysis Help: What's New

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What's New in This Release

The *Synopsys Release Notes* provide information on new features, enhancements, changes, known issues, and resolved Synopsys Technical Action Requests (STARs). To view the release notes, go to [My Product Releases](#) on SolvNetPlus, then choose the product and release version of interest.

New Features, Enhancements, and Changes

This release includes the following new features, enhancements, and changes.

- [PrimeSim CCK: Added or Updated Commands](#)
 - [PrimeSim EMIR: New Features and Enhancements](#)
 - [PrimeSim EMIR: Added or Updated Commands](#)
 - [PrimeSim EMIR: Documentation Updates](#)
 - [PrimeSim Custom Fault: New Features and Enhancements](#)
 - [PrimeSim Design Robustness: New Feature and Enhancements](#)
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PrimeSim CCK: Added or Updated Commands

This release adds or enhances the following commands:

- In the `command`, for the `-proc` option, added the set of pm utility that can be used in the callback procedures.
- In the `,` added the following:
 - `-vt_drop`: Added new value 2 and updated the vt_drop propagation table.
 - `-vt_drop_file file_name`: This new option, sets `vpth` and `vnth` values based on different model name.

For more information, see the *PrimeSim™ CCK Reference Manual*.

PrimeSim EMIR: New Features and Enhancements

The PrimeSim EMIR tool adds the following enhancement:

- New consolidated interface to setup and run flow based simulator agnostic EMIR. Supports Library EMIR, Dynamic EMIR, Fast IR, Static IR Drop and Static Res Check Analyses. The new Results Viewer integrated in the same interface provides Net based violation debugging with multiple heat map support including Pin Current/Voltage Map. The PrimeSim EMIR GUI application continues to support advanced debugging features like What-If Analysis and Least Resistive Path Tracing

For more information, see

The PrimeSim EMIR adds the following enhancement:

- PrimeSim EMIR supports distributed processing at phase two step in EMIR simulation. Now, you can launch distributed processing using command line.

For more information, see Running PrimeSim EMIR.

PrimeSim EMIR: Added or Updated Commands

The PrimeSim EMIR adds or enhances the following commands:

- New commands `set_ra_option -select_subckt` and `set_ra_option -select_inst` are introduced to select one subcircuit or instance in emir analysis if one subcircuit is instantiated more than one time. For more information, see `set_ra_option`.
- New flow option in command line `primesim_emir -flow static_ir` is introduced to support static IR drop Electromigration (IREM) analysis in flow manager based EMIR simulation.

For more information, see Flow Options in EMIR Flow Manager.

- New command `emiroptions` is introduced to define machine readable output files.

For more information, see `emiroptions`.

- The command `genextrview` is replaced with the `genspfview` command. The `genspfview` command provides predefined values for `-cell`, `-view`, and `-lib` OUTPUT Prefix.

For more information, see `genspfview`.

PrimeSim EMIR: Documentation Updates

This release updates the *PrimeSim™ EMIR Reference Manual* as follows:

- Introduced a new section Running EMIR analysis static_ir flow.

PrimeSim Custom Fault: New Features and Enhancements

The PrimeSim Custom Fault application does not have any new feature in this release.

PrimeSim Design Robustness: New Feature and Enhancements

The Design Insights which is part of PrimeSim Design Robustness adds the following enhancements:

- Design Insights analysis automatic mode enhancement - identify pull-up or pull-down paths by sorting the P or N type drain or source terms together and all the gate terms together. In addition, the tool also identifies the paths such as InstPin to port, port to gate, and so on.

For more information, see Advanced Design Exploration With Design Insights.